

WEEK 8 – STUDENT ACADEMIC PERFORMANCE ANALYSIS USING POWERBI

Introduction.

Microsoft Power BI is a powerful data visualization tool used to analyse data and present information through interactive dashboards.

In this Week 8 activity, I developed a Student Academic Performance Dashboard using Power BI. The dashboard helps to understand students' academic results in Mathematics, Reading, and Writing. It uses KPI cards, charts, and slicers to display the information in a clear and organized manner.

The analysis also focuses on different factors, including race/ethnicity, parental education, lunch category, test preparation course, and gender. These visualizations make it easier to compare student performance across different categories.

Objectives.

The major objectives of this activity are:

- *To create an interactive student performance dashboard using Power BI.
- *To calculate and display average scores using KPI cards.
- *To examine Mathematics performance across race/ethnicity groups.
- *To study average scores based on parental education levels.
- *To compare academic results between Standard and Free/Reduced lunch categories.
- *To analyse subject scores based on test preparation course completion.
- * To compare the academic performance of male and female students.
- *To use slicers for interactive data filtering.
- *To represent academic information through meaningful visualizations.

Dashboard Description.

*The Student Academic Performance Dashboard presents important information about student results in a visual format. It contains five KPI cards, multiple charts, and a Race/Ethnicity slicer.

*The KPI cards summarize the main performance indicators, while the charts help to identify differences between various student groups. The slicer allows users to select specific categories and examine the corresponding results.

*This dashboard makes the analysis process simpler and reduces the need to examine individual records manually.

Dashboard Components.

Key Performance Indicator (KPI) Cards

Five KPI cards are used to display the overall academic performance of the students.

Average Mathematics Score.

*The average Mathematics score is 66.09. This value represents the average marks obtained by students in Mathematics and provides an overview of their performance in this subject.

Average Reading Score.

The average Reading score is 69.17. It indicates the average performance of students in Reading and helps evaluate their reading-related results.

Average Writing Score.

The average Writing score is 68.05. This indicator summarizes the students' average marks in Writing.

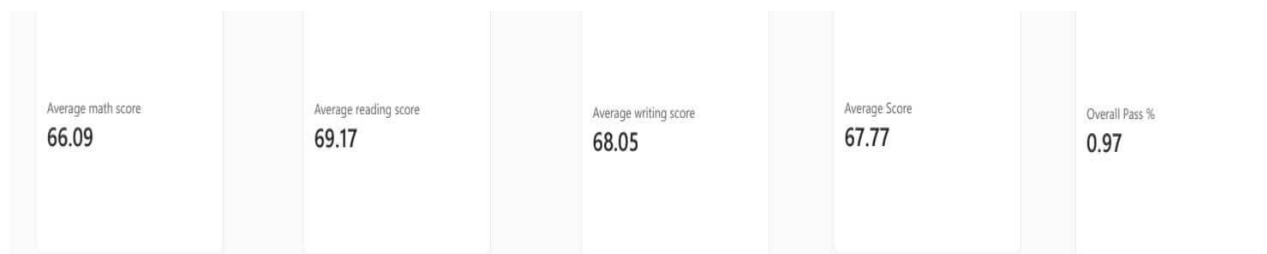
Average Overall Score.

The overall average score is 67.77. It provides a general summary of student achievement across the subjects considered in the dashboard.

Overall Pass Rate.

*The Overall Pass indicator displays 0.97. It presents the overall passing measure shown in the dashboard.

*Together, these KPI cards provide a quick understanding of the major academic performance indicators.



Mathematics Performance by Race/Ethnicity.

This chart compares the average Mathematics marks of five race/ethnicity groups: Group A, Group B, Group C, Group D, and Group E.

The displayed average scores are:

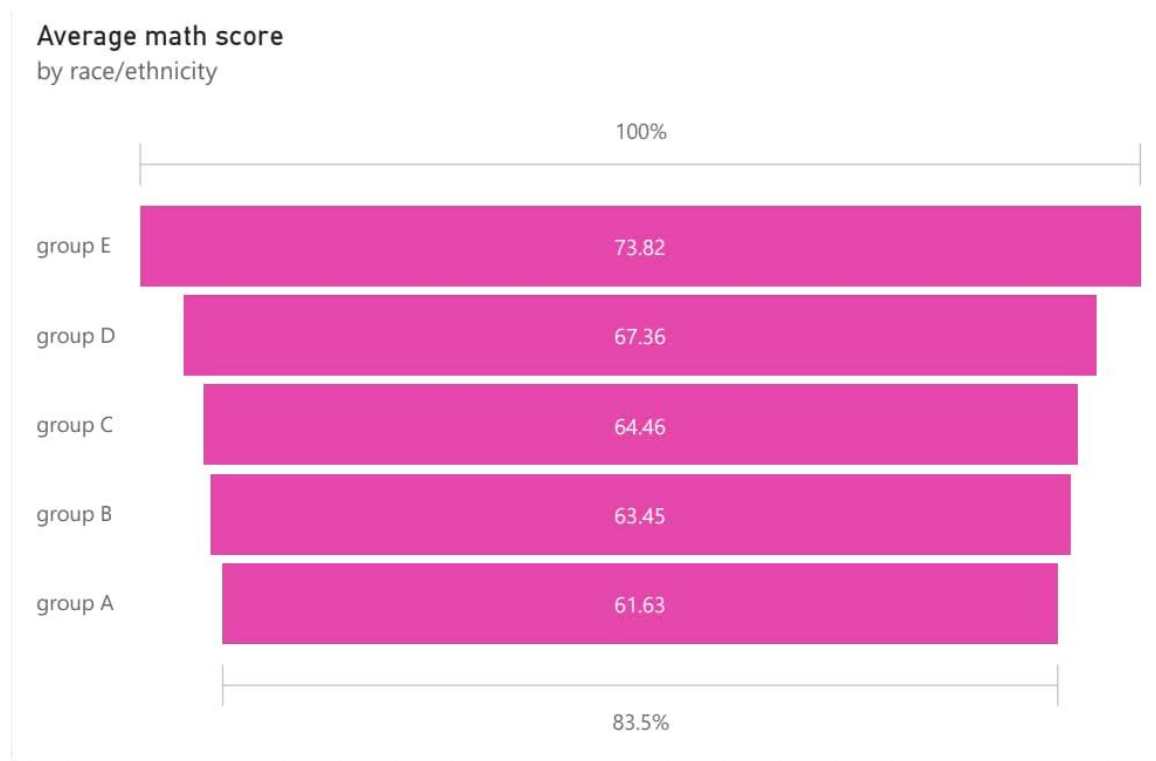
***Group A – 61.63.**

***Group B – 63.45.**

***Group C – 64.46.**

***Group D – 67.36.**

***Group E – 73.82.**

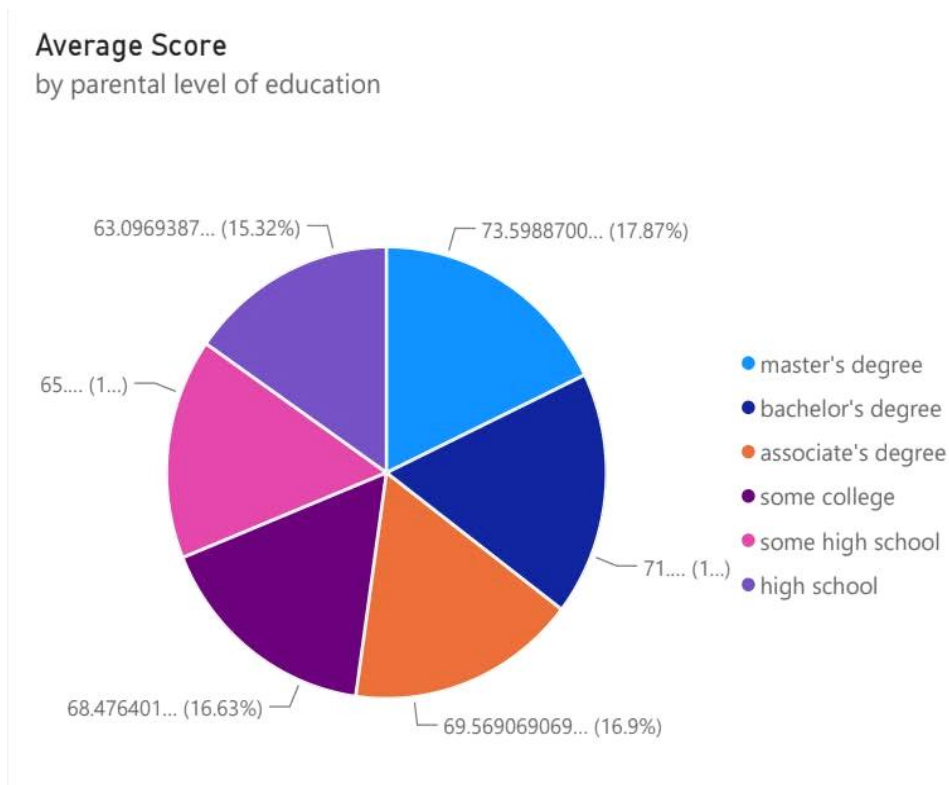


Average Student Score by Parental Education.

This visualization examines student scores according to their parents' educational qualifications.

The categories included are:

- *Master's degree.
- *Bachelor's degree.
- *Associate's degree.
- *Some college.
- *Some high school.
- *High school.



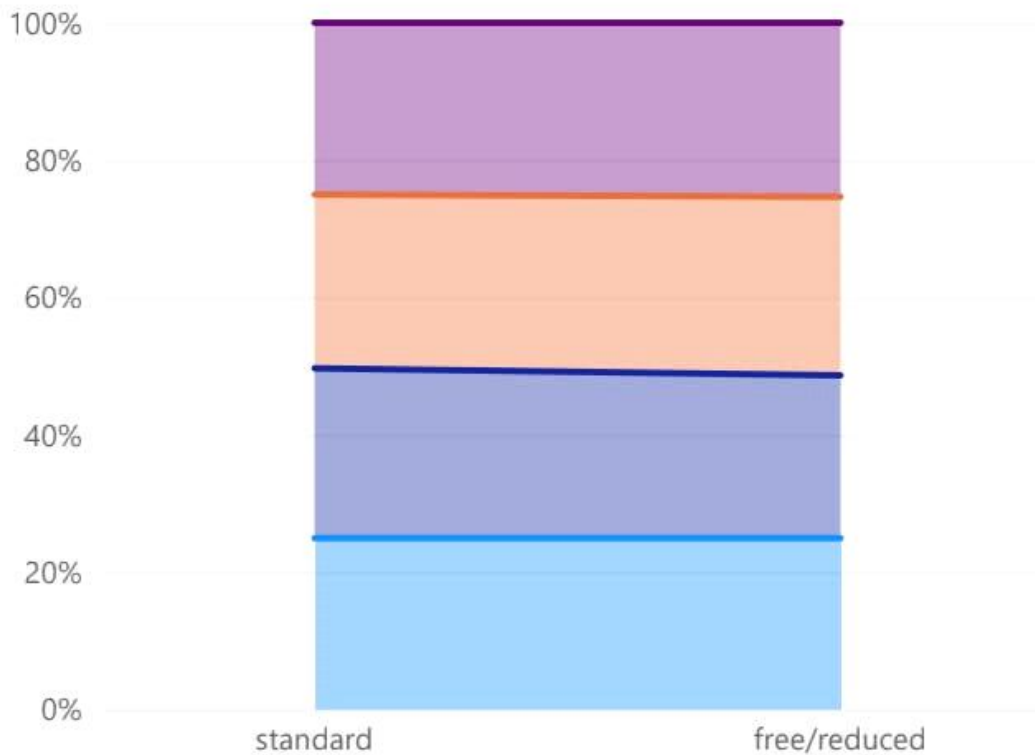
Academic Performance Based on Lunch Category.

This chart compares students belonging to the Standard and Free/Reduced lunch categories.

The following measures are included:

- *Average Overall Score
- *Average Mathematics Score
- *Average Reading Score
- *Average Writing Score

Average Score, Average math score, Average reading score
and Average writing score
by lunch



Subject Scores by Test Preparation Course.

This visualization analyses students according to their test preparation course status.

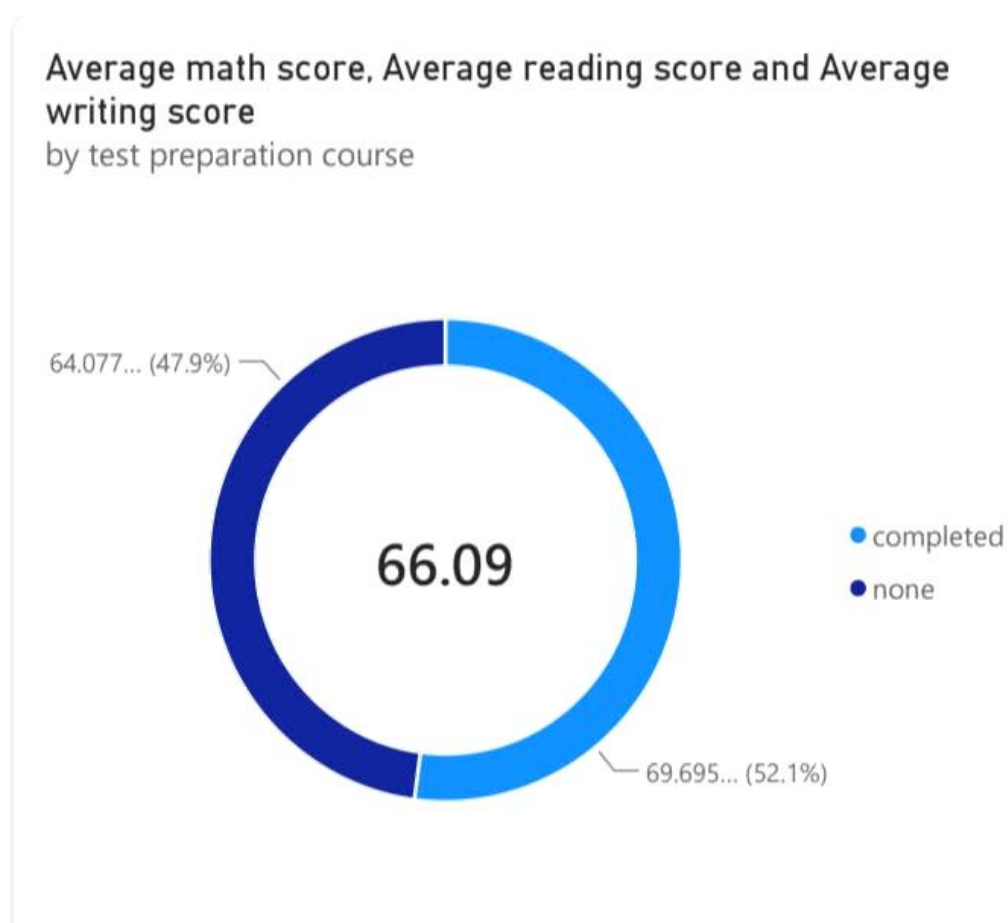
The two categories are:

***Completed.**

***None.**

The average Mathematics, Reading, and Writing scores are compared between these categories. This helps users observe the differences in subject performance between students who completed the preparation course and those who did not.

The chart provides a visual summary of academic performance according to test preparation status.



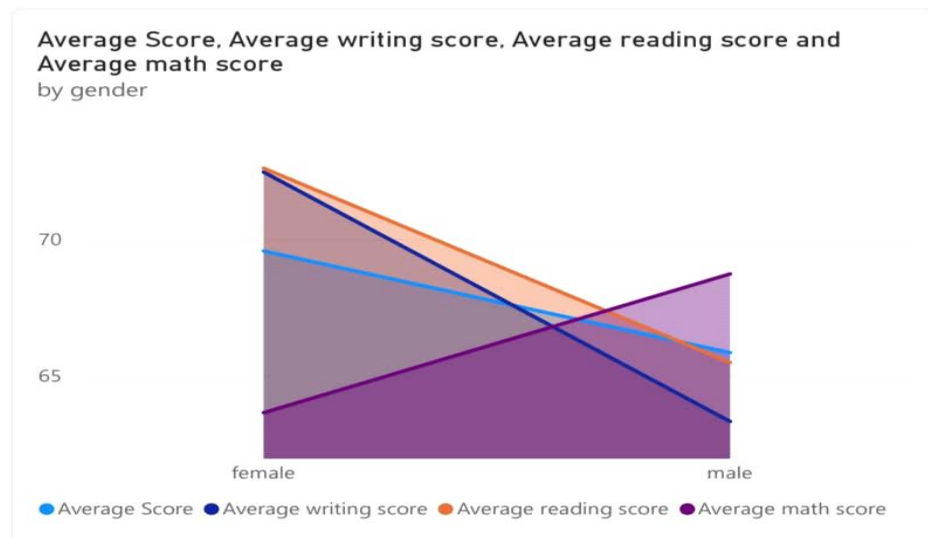
Academic Performance by Gender.

This chart compares the average academic results of Female and Male students.

The measures displayed include:

- *Average Overall Score
- *Average Writing Score
- *Average Reading Score
- *Average Mathematics Score

The visualization presents the overall and individual subject averages for both categories. It helps users identify differences in the displayed scores and understand the distribution of academic performance by gender.



Subject-Wise Comparison by Lunch Category.

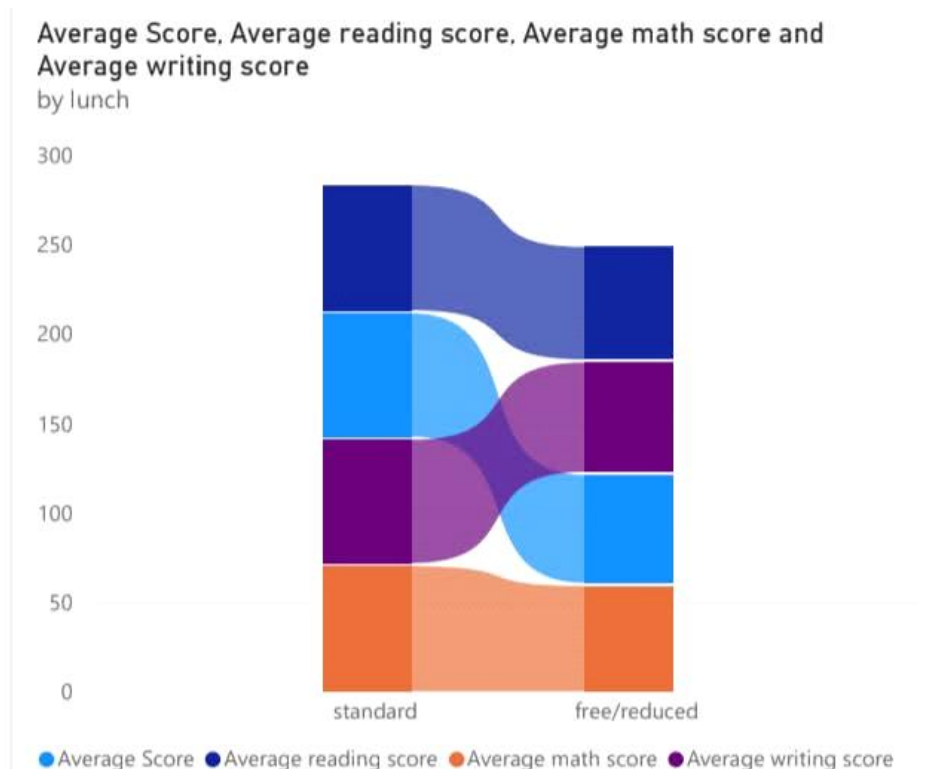
This visualization provides another comparison of subject performance between Standard and Free/Reduced lunch categories.

It displays the following measures:

- *Average Overall Score
- *Average Reading Score
- *Average Mathematics Score
- *Average Writing Score

By showing all four indicators together, the chart makes it easier to examine

the academic results of both groups. It also supports a more detailed comparison of overall performance and individual subject scores.



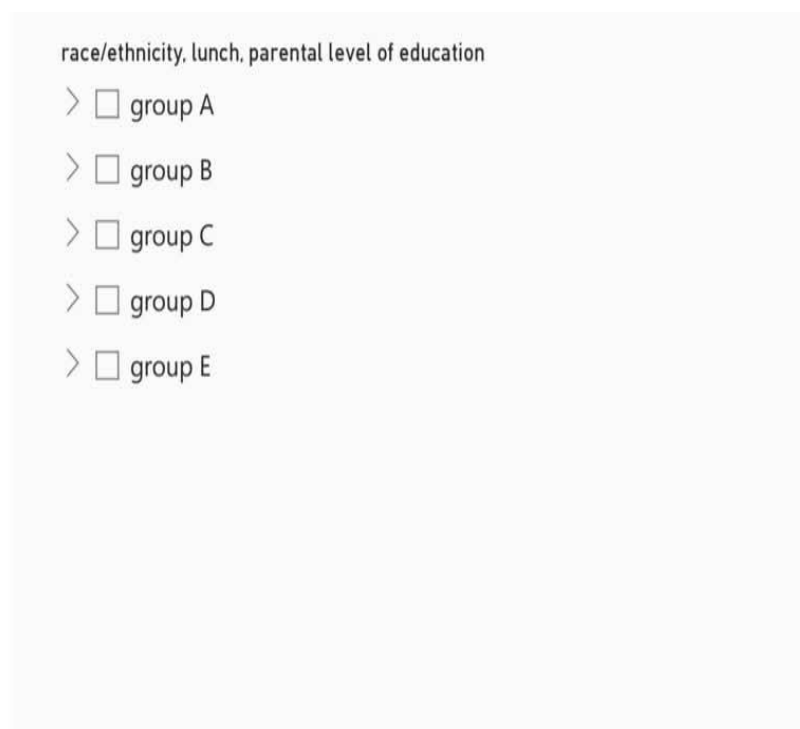
Race/Ethnicity Slicer.

- * A Race/Ethnicity slicer is included to improve the interactivity of the dashboard.

- * The slicer contains Group A, Group B, Group C, Group D, and Group E. Users can select a particular group to filter the

dashboard and focus on the relevant student records.

* This feature makes the dashboard more flexible and helps users analyse specific groups without manually searching through the dataset.



Key Findings.

The Student Performance Dashboard provides several useful observations about the academic data.

*The average Mathematics score is 66.09.

- *The average Reading score is 69.17.
- *The average Writing score is 68.05.
- *The overall average score is 67.77.
- *Among the five race/ethnicity groups, Group E records the highest displayed average Mathematics score of 73.82.
- *Group A records the lowest displayed average Mathematics score of 61.63.
- *The parental education chart supports comparisons of student averages across six educational categories.
- *The lunch category chart allows users to compare overall and subject-wise scores between Standard and Free/Reduced groups.
- *The test preparation chart compares the displayed subject averages for students who completed the course and those who did not.

*The gender chart provides a comparison of academic performance between Female and Male students.

Overall, the dashboard organizes student information into different visual categories, making the data easier to interpret and compare.

Conclusion.

The Week 8 Student Academic Performance Dashboard was developed to present student results in an organized and interactive manner using Microsoft Power BI.

The dashboard combines KPI cards, comparison charts, and a Race/Ethnicity slicer to examine Mathematics, Reading, and Writing scores. It also supports comparisons based on parental education, lunch category, test preparation course, and gender.

The visualizations help users understand the main performance indicators and explore differences among the various student groups. The slicer further improves the dashboard by allowing users to filter the displayed information according to their needs.

In conclusion, this activity demonstrated how Power BI can transform student academic data into meaningful visual information and improve the presentation, comparison, and interpretation of data.

Submitted By:

Sivaranjani A.